

LAN Subject Type A.

T1). Does Gigabit Ethernet require a longer minimum length of the transmission frame than the 10/100 Mbps Ethernet? If so, what is the minimum frame length? (0,5p)

T2). a). Describe the topology of a FDDI network.

b). Enumerate the types of FDDI stations.

c). Enumerate the transmission media allowed to be used in a FDDI network. (1p)

T3). Present the IPv6 protocol: functions, reliability, addressing, header structure, option headers (1,5p)

T4). The DNS application service (aim, possible implementations, DNS name structure) (1p)

T5). a). Differentiate between message-switched and circuit-switched networks. (0,5p)

b). Explain the basic reason why packet switching is more suited for data communications.

T6). Internet Protocol has a Time-To-Live field in its header, which essentially counts hops. (0,5p)

(a). What is its purpose, and why is it needed?

(b). Identify specific reasons that might cause the conditions that the TTL field protects against.

T7). Present the main functionalities of the TCP protocol (based on the header structure). (1p)

T8). Which are the light propagation modes through a fiber? (1p)

P1). Consider the use of 1000 bit frames on a 1Mbps satellite channel with a 270ms delay.

What is the maximum link utilization for:

a). Stop-and-wait protocol?

b). Continuous flow control with a window size of 127 frames? (1p)

P2). The IP address on a host computer is 129.6.31.101 and the subnet mask is 255.255.255.192.

(a) What class of IP address is 129.6.31.101?

(b) After applying the subnet mask, what is the structure of this address, that is, how many bits are assigned to net, subnet, and host identifiers?

(c) If the host desires to broadcast to all hosts on the same network and subnet as itself, then what destination IP address should the host choose? (2p)